

```

    outtextxy(260,160,"PLAY GAME
SCREEN");
    settextrstyle(0, 0, 0);
    setcolor(03);
    draw_box(es);
    vplay();
    shuffled_flag(s);
}
void demo_play()
{
    gameinst();
    init_box(es);
    shuffle_box(es,dnum);
    settextrstyle(0, 0, 01);
    setcolor(04);
    outtextxy(280,150," DEMO");
    outtextxy(260,160,"PLAY GAME
SCREEN");
    settextrstyle(0, 0, 0);
    setcolor(03);
    draw_box(es);
    vdemo();
    shuffled_flag(s);
}
void hard_play()
{
    gameinst();
    init_box(es);
    shuffle_box(es,hnum);
    settextrstyle(0, 0, 01);
    setcolor(04);
    outtextxy(280,150," GIVEN");
    outtextxy(260,160,"PLAY GAME
SCREEN");
    settextrstyle(0, 0, 0);
    setcolor(03);
    draw_box(es);
    vplay();
    shuffled_flag(s);
}
void init_box(es[square sq[]])
{
    for(int i = 0; i < 4; i++)
    {
        for(int j = 0; j < 4; j++)
        {
            sq[4*i+j].left = fleft+sqxlen*j;
            sq[4*i+j].top = ftop+syglen*i;
            sq[4*i+j].right = fleft+sqxlen*j+
            sqxlen;
            sq[4*i+j].bottom = ftop+syglen*i+
            syglen;
            sq[4*i+j].value = num[4*i+j];
            sq[4*i+j].valflag = 4*i+j;
        }
    }
    void draw_box(es[square sq[]])
    {
        int gap = 5, outl = 15;
        rectangle(fleft-5, ftop-5, fleft+4*
        (5+sqxlen), ftop+4*(5+syglen));
        rectangle(fleft-7, ftop-7, fleft+4*
        (7+sqxlen)-5, ftop+4*(7+syglen)-5);
        settextrstyle(0, 0, 0);
        settextrstyle(01,63);
        for(int i = 0; i < 16; i++)
        {
            setcolor(2);
            rectangle(sq[i].left+gap*2, sq[i].
            top+gap*2, sq[i].right+gap, sq[i].
            bottom+gap);
            if(atoi(sq[i].value) == 0) //i=15
            { settextrstyle(01,02); outl = 20; c=i*4;
            r = 1/4; }
            else
            { settextrstyle(01,63); outl = 15; }
            bar(sq[i].left+15, sq[i].top+15,
            sq[i].right, sq[i].bottom);
            setcolor(60);
            outtextxy(sq[i].left+outl, sq[i].
            top+20, sq[i].value);
        }
        void shuffle_box(es[square sq[]], char
        *snum[])
        {
            for(int i = 0; i < 4; i++)
            {
                for(int j = 0; j < 4; j++)
                {
                    sq[4*i+j].value = snum[4*i+j];
                    if(atoi(snum[4*i+j]) == 0)
                    sq[4*i+j].valflag = 15;
                }
            }
            sq[4*i+j].valflag = atoi(snum[4*i+j])-
            1;
        }
        void vdemo()
        {
            int temp, i, j = 0;
            dis_rem_moves();
            for(int k = 0; k < 20; k = k +
            2)
            {
                dis_rem_moves();
                vbell();
                i = dloc[k];
                getimage(s[i].left+15,
                s[i].top+15, s[i].right, s[i].
                bottom, vbuff1);
                delay(500);
                k++;
                i = dloc[k];
                getimage(s[i].left+15,
                s[i].top+15, s[i].right, s[i].
                bottom, vbuff2);
                delay(500);
                putimage(s[i].left+15, s[i].
                top+15, vbuff1, 0);
                j++;
                temp[s[i].valflag]; /* copy location
                of i flag image */
                k--;
                i = dloc[k];
                putimage(s[i].left+15, s[i].
                top+15, vbuff2, 0);
                s[i].valflag = temp; /* copy loca-
                tion of i flag image */
                j++;
                vbell();
                settextrstyle(0, 0, 02);
                outtextxy(180,400,"YOU WON THE
                GAME !");
                settextrstyle(0, 0, 01);
                settextrstyle(0, 0, 0);
                setcolor(04);
                outtextxy(200,360," <Press any
                key to return to main >");
                getch();
            }
            void dis_rem_moves()
            {
                char n[10];
                ++jmoves;
                rectangle(450,190,600,240);
                outtextxy(460,200,"Moves:
                50");
                outtextxy(460,210,"Remaining
                Moves:");
                itoa(50-jmoves,n,10);
                setviewport(510,220,540,
                230,0);
                clearviewport();
                setviewport(510,220,540,
                230,0);
            }
            outtext(n);
            setviewport(0,0,getmaxx(),getm
            axy(),0);
        }
        int getkey()
        {
            union REGS in, out;
            int sc;
            in.h.ah = 00;
            int86(0x16, &in, &out);
            sc = out.h.ah;
            return(sc);
        }
        void vplay()
        {
            int i, temp, k;
            jmoves = 0;
            while((key = getkey()) != 1)
            {
                switch(key)
                {
                    case 72 : /*up arrow */
                        dis_rem_moves();
                        if(r <= 0)
                        {
                            /* display error message */
                            setviewport(200,350,550,
                            360,0);
                            outtext("Not allowed use other
                            than up arrow");
                            vbell();
                            delay(200);
                            clearviewport();
                            setviewport(0,0,getmaxx(),getm
                            axy(),0);
                        }
                        break;
                        /* other processing */
                        --r;
                        i = 4*r+c;
                        temp[s[i].valflag];
                        k = i;
                        getimage(s[i].left+15, s[i].
                        top+15, s[i].right, s[i].bottom,
                        vbuff1);
                        ++r;
                        i = 4*r+c;
                        s[k].valflag = s[i].valflag;
                        s[i].valflag = temp; /* copy loca-
                        tion of i flag image */
                        getimage(s[i].left+15, s[i].
                        top+15, s[i].right, s[i].bottom,
                        vbuff2);
                        putimage(s[i].left+15, s[i].
                        top+15, vbuff1, 0);
                        --r;
                        i = 4*r+c;
                        putimage(s[i].left+15, s[i].
                        top+15, vbuff2, 0);
                        break;
                        case 80 : /*down arrow */
                            dis_rem_moves();
                            if(r >= 3)
                            {
                                setviewport(200,350,550,
                                360,0);
                                outtext("Not allowed use other
                                than down arrow");
                                vbell();
                                delay(200);
                                clearviewport();
                                setviewport(0,0,getmaxx(),getm
                                axy(),0);
                            }
                            break;
                            /* other processing */
                            ++r;
                }
            }
        }
    }
}

```